Technical Guide: Hydraulic Press System Troubleshooting

Hydraulic Circuit Diagnostics, Proportional Valves & Heat Exchangers

ASSET CODE: HYD-PRESS-P14 **DOC REF:** HYD-PRESS-MAN-2026 **STANDARD:** ISO 4413 **BAY:** Heavy Stamping Bay

1. Operating Envelope & Threshold Limits

Hydraulic Stamping Press **HYD-PRESS-P14** operates on an electro-hydraulic closed circuit with high-precision directional proportional valves:

- **Main Operating Pressure:** Nominal 160 bar (Safety pressure relief valve set at 185 bar).
- **Operating Fluid Temperature:** Nominal 40 - 50°C. Maximum continuous alarm at 55°C.
- **Directional Valve Response Time:** Nominal 42 - 45 ms. Alarm triggered if switching latency exceeds 48 ms.
- **Accumulator Pre-charge Pressure:** Pure nitrogen nominal 70 bar in bladder unit `ACC-01`.

2. Fault Symptoms & Corrective Maintenance SOPs

A. Reduced Cooling Circuit Efficiency (< 40%)

Root Cause: Mineral calcification or biofilm silt on secondary cooling plates; jammed thermostatic bypass valve.
Corrective Action: Isolate plate heat exchanger. Flush secondary circuit with 10% citric acid descaling agent at 45°C for 120 minutes. Flush with demineralized water until neutral pH is confirmed.

B. Proportional Valve Response Lag (> 48 ms)

Root Cause: Silt deposits in proportional spool feedback pilot; drop in accumulator nitrogen charge.
Corrective Action: Measure nitrogen pre-charge on accumulator `ACC-01` using testing manifold. Recharge to 70 bar dry N_2 if pressure < 50 bar. Replace 3-micron inline servo filter cartridge.

C. Excessive Internal Pump Leakage

Root Cause: Slipper pad cavitation wear on axial piston pump barrel due to low fluid viscosity or contamination.
Corrective Action: Draw oil sample for ISO 4406 cleanliness count (Target: 16/14/11). Drain and replace ISO VG 46 hydraulic oil if kinematic viscosity drops below 40 cSt at 40°C.